

What Makes a GEMASTIK Cyber Winner

Executive summary

A GEMASTIK Cyber winner is not simply “the best Jeopardy player.” Across 2023, 2024, and 2025, the official rules show a stable pattern: the qualifier is a Jeopardy-style CTF with broad topic coverage, but the final is a live Attack-Defense round followed by a write-up presentation. That means the winning profile is a hybrid one: strong breadth during qualifiers, then fast exploit automation, disciplined patching, stable service operation, and clear technical communication during finals. The official guides also keep the finalist pool at 20 teams, keep a live scoreboard, require formal write-up submission, and award special non-medal distinctions such as Best Defender and Best Attacker in Cyber Security. ①

The most reliable competitive signal in recent years is institutional continuity. Official rankings and campus announcements show recurring strength from Universitas Indonesia ②, Universitas Bina Nusantara ③, Institut Teknologi Bandung ④, Institut Teknologi Sepuluh Nopember ⑤, and Universitas Gadjah Mada ⑥. That pattern strongly suggests that winning GEMASTIK Cyber is less about one-off brilliance and more about a repeatable system: internal selection, category specialization, coach/alumni review, scrimmages, automation habits, and final-week operational discipline. ⑦

The official public rules are detailed on format, timelines, and topic families, but they do not publish a complete, official, cross-year numeric dump of every challenge’s point values. The public evidence supports a rigorous analysis of the format, recurrent topic families, winners, finalist ecosystems, and public write-up patterns; exact cross-year per-category point histograms are only partially recoverable from public archives. Where the public record is incomplete, this report says so directly rather than inventing precision. ⑧

Official format and scoring

The official guides for 2023, 2024, and 2025 define a consistent structure for Cyber Security. The qualifier is a Capture the Flag round in Jeopardy format. Competitors receive standalone challenges, submit flags to the platform, and see a live scoreboard. Each challenge has a point value or “weight” that varies by difficulty, and teams must also submit a write-up after the round. The published qualifier topic families are stable across all three years: system hardening, web, digital forensics, cryptography, binary analysis, steganography, reverse engineering, and network sniffing. ⑨

The final is fundamentally different. In every recent edition, the official rules describe a two-part final: an offline Attack-Defense competition followed by a presentation of the team’s write-up on the next day. In the Attack-Defense phase, each team receives a vulnerable system, must patch and maintain its own service, and then attack opponents to capture flags while protecting its own. The rules explicitly prohibit DoS, excess damage, flag sharing, and collaboration outside the team. This means final-round success is not only exploit skill; it is operational security, service reliability, and controlled aggression. ⑩

A crucial but often overlooked scoring implication is that GEMASTIK Cyber finals are not “scoreboard only.” The official schedules require same-day write-up submission and next-day presentation. The 2024 final handbook schedules an Attack-Defense session, then overnight write-up time, and then formal presentation slots for all 20 teams; the 2025 guide keeps the same shape, with the competition on one day and presentation on the following day. In practice, this rewards teams that can preserve evidence, document quickly, and explain both offense and defense under time pressure. ⑪

Jeopardy and Attack-Defense are testing different things

Dimension	Jeopardy qualifier	Attack-Defense final
Primary question	“Can you solve many diverse problems?”	“Can you keep a vulnerable service alive while exploiting others?”

Work unit	Mostly independent challenge instances	Shared live services under time pressure
Scoring style	Flag solves, challenge points vary by difficulty, live scoreboard	Live scoreboard plus operational success, then write-up presentation
Team advantage	Breadth across categories	Specialization + orchestration + automation
Typical failure mode	Spending too long on rabbit holes	Breaking your own service, slow patching, non-automated flag stealing
What wins	Broad solve rate and selective deep solves	Fast patch loop, exploit replayability, clean roles, evidence retention

The table above is a direct consequence of the official rules, schedules, and public finalist-write-up materials. ¹²

What the score structure implies for preparation

Because the qualifier spans many topic families while the final concentrates on live exploitation and defense, the training implication is straightforward: a serious medal contender must build two engines. The first engine is a breadth engine for qualifier consistency. The second is an operations engine for the final: exploit automation, patch discipline, local deployment, service monitoring, and rapid write-up production. Teams that train only for Jeopardy tend to underperform once the format shifts to Attack-Defense. ¹³

Winners, finalists, and what their profiles imply

The official ranking pages show three different Cyber champions in 2023–2025: team apanii from Universitas Gadjah Mada ⁶ in 2023, PETIR - FlagGPT from Universitas Bina Nusantara ³ in 2024, and Dafi Nafidz Radhiyya from Universitas Indonesia ² in 2025. The runner-up and bronze positions, however, cluster around a relatively small set of institutions, especially UI, BINUS, ITB, ITS, UGM, UB, and UNHAS. That concentration strongly suggests the value of a durable campus-level training culture rather than individual talent alone. ¹⁴

Medalists and selected profiles

Year	Gold	Silver	Bronze	Short profile	Official links
2023	apanii — UGM	Moai — BINUS	Calon Mantune Bapakmu — UMS	2023's official ranking shows a UGM win, with BINUS and UMS also on the podium. Official rules and campus reporting from the same year already show that the final emphasized Attack-Defense and special defensive recognition.	; ; ¹⁵
2024	PETIR - FlagGPT — BINUS	kessoku no owari — ITB	AcRtf — UB	2024's official ranking confirms BINUS gold, ITB silver, and UB bronze. Public finalist reporting shows the ecosystem around these teams was already highly competitive and nationally visible.	; ¹⁶
2025	Dafi Nafidz Radhiyya — UI	PETIR mencari cici cici daerah — BINUS	ICC Pisang Molen — UNHAS	UI's official campus article names the gold-medal team	; ; ; ; ²⁰

				members as Steven Setiawan ²⁰ , Dimas Herjunodarpito Notoprayitno ²¹ , and Bimantoro Widyadana ²² . BINUS's official article lists the silver team members, and UNHAS reporting identifies the bronze team members. UI also publicly reported the Best Defender award for team "ijazahnya mana woy."	
--	--	--	--	---	--

A practical reading of these results is that recent GEMASTIK Cyber winners usually emerge from teams with at least three characteristics: prior CTF exposure outside GEMASTIK, internal campus mentoring or alumni continuity, and an institution that already knows how to prepare students for the final's Attack-Defense shift. The 2025 articles on UI, BINUS, and UNHAS are especially revealing because they show the medalist teams not as isolated individuals but as identifiable campus cyber groups with continuity and coaching. ²³

For completeness, the official finalist rosters are publicly available in the linked 2023, 2024, and 2025 finalist announcements. Those rosters matter because GEMASTIK Cyber is increasingly deep: reaching the final is already a national-level signal, and the gap between a lower finalist and a medalist is often process quality rather than raw skill. ²²

Challenge coverage, category recurrence, and the limits of public point data

The strongest cross-year empirical result that can be established from official sources is category stability. The official guides for 2023, 2024, and 2025 repeat the same qualifier topic families: system hardening, web, digital forensics, cryptography, binary analysis, steganography, reverse engineering, and network sniffing. The final, in all three years, remains Attack-Defense plus presentation. For a team preparing to win, this is excellent news: the competition is broad, but it is not unpredictable. ²³

What is not fully public is exact per-challenge numeric point distribution for every year. The official rules say challenge points differ by difficulty, but they do not publish a complete cross-year numeric challenge list in the rules themselves. Public write-ups and repos reveal many individual tasks, but not enough to claim a complete official histogram without overreaching. The rigorous conclusion is therefore this: you should trust the official topic recurrence, but you should not assume a fully public, exact, official cross-year point spreadsheet exists for 2023–2025. ²⁴

Officially recurring topic families

Topic family	2023	2024	2025	Practical mapping to training
System hardening	Yes	Yes	Yes	Service configuration, patching, permissions, auth hardening
Web exploitation	Yes	Yes	Yes	XSS, SSRF, auth flaws, injections, logic bugs
Digital forensics	Yes	Yes	Yes	File artifacts, logs, browser traces, memory-style reasoning
Cryptography	Yes	Yes	Yes	AES modes, RSA, ECDSA, nonce issues, encoding chains
Binary analysis	Yes	Yes	Yes	pwn foundations, ELF analysis, memory bugs

Steganography	Yes	Yes	Yes	Hidden data in media; often operationally part of forensics
Reverse engineering	Yes	Yes	Yes	Static/dynamic analysis, decompilation, custom VMs
Network sniffing	Yes	Yes	Yes	PCAPs, protocol reversal, traffic triage
Attack-Defense final	Yes	Yes	Yes	Patch, monitor, attack, document, present

This table is derived directly from the official published guides. 23

Public archive observations

The public archive confirms that these official categories are not just nominal. A 2023 original crypto write-up explicitly states that the 2023 qualifier had three cryptography challenges; a 2024 public write-up covers at least web and forensics tasks such as Baby XSS and Baby Structured; another 2024 public archive notes solves across crypto, forensics, reverse engineering, and web; and a 2025 medalist's public post describes the 2025 qualifier as covering web, cryptography, binary exploitation, reverse engineering, and digital forensics, while the final emphasized web, cryptography, and binary exploitation in Attack-Defense. That public evidence is consistent with the official category universe and with the final's operational emphasis. 25

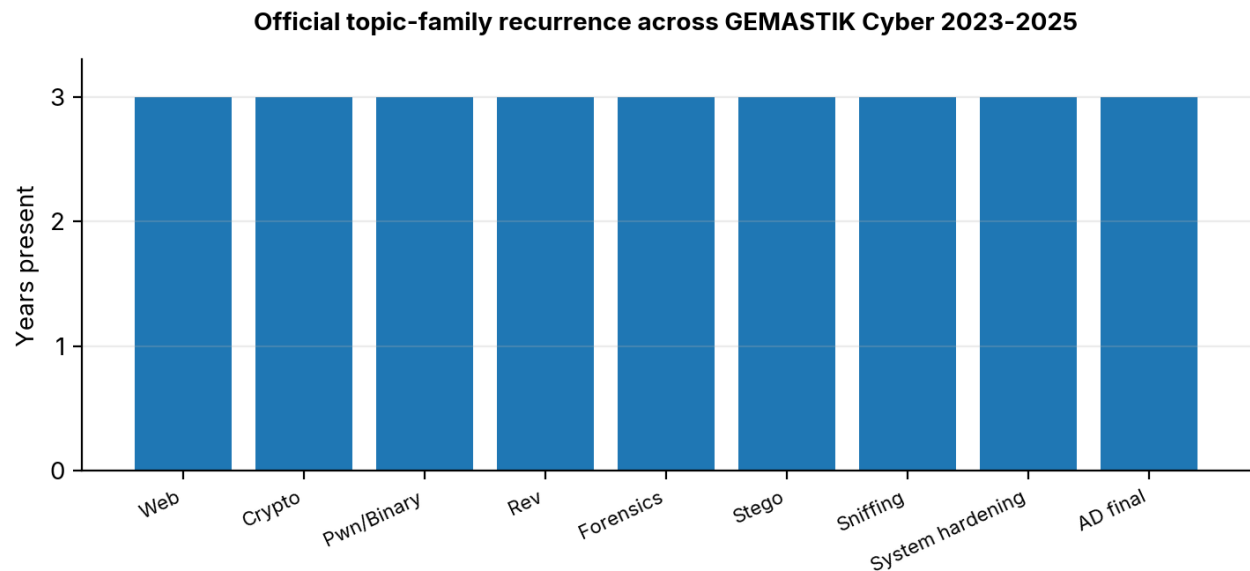


Figure 1. Official topic-family recurrence across GEMASTIK Cyber 2023-2025.

The chart above measures official recurrence, not hidden platform point totals. That is the most defensible public metric available across all three years. 26

GEMASTIK Cyber topic continuity

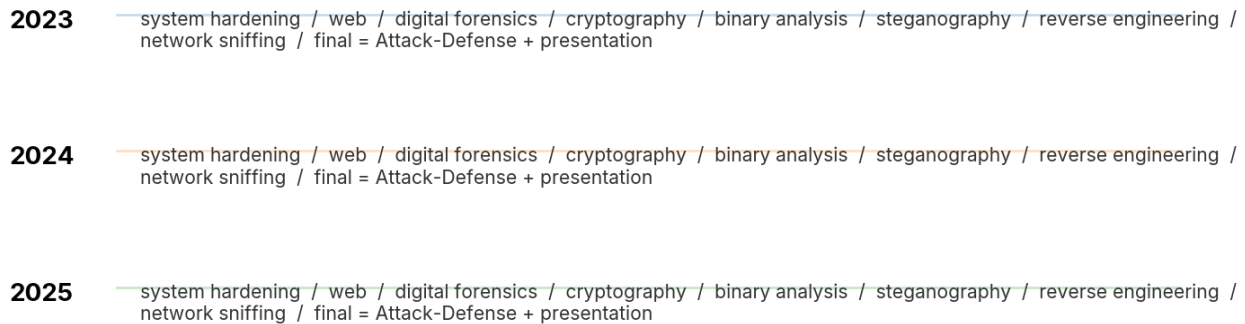


Figure 2. GEMASTIK Cyber topic continuity, 2023-2025.

Skills, tools, and winner workflows

A winning GEMASTIK team needs category depth, but it also needs a category conversion strategy: how to turn partial understanding into solves quickly enough to survive Jeopardy, and then how to turn a foothold into repeated exploit/patch cycles in the final. Public write-ups from recent finals are especially useful here because they show what finalists actually automate and document. The 2023 burvesigner final write-up explicitly states that an Attack-Defense challenge had to be exploitable with an automation script within one tick. The 2024 ASMR final gist shows an aggressive pwntools-based exploit chain. The 2024 FJB write-up shows the value of backend-first reasoning when frontend code is incomplete. These are exactly the habits GEMASTIK rewards. ²⁷

Category-by-category priority map

Category	Prioritized learning topics	Core tools	Example exercises
Web	HTTP internals, cookies/sessions, authZ, SQLi, XSS, SSTI, SSRF, deserialization, file upload, basic JS and browser behavior	Burp Suite Community or Pro, ffuf, httpx, jq, a local browser devtools workflow	especially SQLi, XSS, SSRF, auth, and request smuggling paths ²⁸
Crypto	Modular arithmetic, AES modes, XOR reasoning, RSA, signatures, ECDSA nonce failures, PRNG issues, encoding pipelines	Python, SageMath, PyCryptodome, pwntools helpers	, plus reproducing public GEMASTIK crypto write-ups ²⁹
Pwn	ELF basics, calling convention, canary/NX/PIE/RELRO, ret2libc, ROP, format strings, UAF, heap/tcache, shellcode	pwntools, gdb, gef/pwndbg, checksec, ropper/ROPgadget	; rewrite and replay recent final-style exploit scripts locally ³⁰
Rev	x86-64 assembly, decompilation, strings/XREFs, anti-debug awareness, custom VM reasoning	Ghidra, IDA Free, Binary Ninja if available, radare2, strings, ltrace/strace	Public challenge archives and reverse-focused solve write-ups, especially mixed rev+pwn cases ³¹
Forensics	File formats, carving, metadata, stego, PCAP triage, log analysis, browser artifacts, process/memory	Wireshark, binwalk, exiftool, foremost, Volatility, zsteg, stegsolve	Public GEMASTIK forensic write-ups and network-sniffing cases; classify each artifact source before tooling

	mindset		it ⑫
Attack-Defense	Linux service administration, patch diffing, health checks, container familiarity, exploit replay loops, observability, rollback discipline	tmux, ssh, curl, jq, tcpdump, systemctl, Docker/Compose, supervisor/nginx basics, pwntools/requests loops	Internal scrimmages that force you to patch first, verify uptime, then automate flag theft per tick ⑬

A useful mental model is that qualifier preparation is skill acquisition, while final preparation is skill operationalization. Web, crypto, pwn, rev, and forensics are the feeder disciplines; system hardening and network reasoning are the glue disciplines; Attack-Defense is the stress test that merges them. ⑭

Reusable exploit and defense patterns

A reproducible GEMASTIK-style exploit workflow normally has four stages: initial triage, local reproduction, exploit minimization, and batch automation. Finalists who manually re-exploit a challenge every time usually lose time. The public burvesigner final note is explicit that Attack-Defense challenge design assumes per-tick automation. The public ASMR exploit gist, meanwhile, shows the kind of scriptability a serious finalist uses in practice. ⑮

```
# HTTP target loop skeleton for web/crypto style Attack-Defense services
import requests, time

TARGETS = ["10.10.1.11", "10.10.1.12", "10.10.1.13"]
TEAM_TOKEN = "... "

def health(ip):
    r = requests.get(f"http://{ip}:8080/health", timeout=2)
    return r.status_code == 200

def exploit(ip):
    s = requests.Session()
    # 1) register/login if needed
    # 2) trigger primitive
    # 3) extract flag
    # 4) return normalized flag string
    return "flag{...}"

def submit(flag):
    # local function to push to organizer endpoint
    pass

while True:
    for ip in TARGETS:
        try:
            if health(ip):
                flag = exploit(ip)
                submit(flag)
        except Exception:
            pass
    time.sleep(10)
```

```
# pwntools-style skeleton for pwn services
from pwn import *

HOSTS = ["10.10.1.21", "10.10.1.22"]
PORT = 31337

def get_flag(host):
    io = remote(host, PORT, timeout=2)
    # negotiate, leak, calculate base, ROP/shellcode, read flag
    io.sendline(b"...")
    data = io.recvrepeat(1)
    io.close()
    return extract_flag(data)

for host in HOSTS:
    try:
        print(host, get_flag(host))
    except:
        pass
```

A winner-level defense workflow is just as formalized as the exploit workflow: enumerate attack surface, identify flag-storage paths, patch the narrowest possible vulnerability, verify the service still serves legitimate traffic, add a local health probe, and preserve a diff or revert script. The biggest losing move in Attack-Defense is “fixing” the box by breaking the service. ③⑥

Write-up and presentation workflow

Because GEMASTIK converts the final into a written and oral deliverable, the write-up process should begin during the Attack-Defense round, not after it. The fastest teams keep a rolling evidence log with screenshots, exploit notes, service hashes, patch diffs, and solved-flag provenance. Then the presentation becomes a structured explanation rather than a reconstruction from memory. The official schedules make this mandatory in practice by putting write-up submission and presentation immediately after the live technical round. ③⑦

A strong write-up template has only five sections: problem statement, initial observations, exploit or defense chain, evidence and validation, and final lessons learned. In finals, “what did you patch and why did the service stay alive?” often matters as much as “how did you pop the opponent?” ③⑧

Suggested finals team design

Three-person team

Role	Main ownership	During warm-up	During Attack-Defense	During write-up/presentation
Captain / offensive lead	Web+pwn triage, target prioritization	Verifies tooling, target notes, fallback plan	Drives first blood and exploit replay	Builds narrative, owns opening/closing slides
Defense / infra lead	Patch loop, health checks, logs, rollback	Local service boot tests and health monitors	Patches, validates uptime, watches regressions	Documents defense rationale and service validation
Analyst / scribe	Crypto/rev/forensics support, flag normalization, evidence capture	Creates note templates and timestamped logs	Assists whichever stream is blocked; captures artifacts continuously	Assembles PDF write-up and evidence appendix

This is the minimum viable configuration for a GEMASTIK-winning team, but it is fragile: if one person is weak or overloaded, both offense and documentation can stall. ③⑨

Five-person team

Role	Main ownership	During warm-up	During Attack-Defense	During write-up/presentation
Captain / strategy	Prioritization and timekeeping	Target matrix and comms plan	Chooses which targets and vulnerabilities matter first	Presentation lead
Web attacker	HTTP/auth/SSTI/XSS/ logic bugs	Browser/Burp setup and local HTTP scripts	Repeats reliable web steals	Explains web offense
Pwn attacker	Memory corruption and service exploitation	Toolchain, gdb/pwntools templates	Handles binary roots and exploit stabilization	Explains binary offense
Defender / infra	Patching, checks, logs, restarts	Deployment and verification	Keeps service alive and watches for regressions	Explains defense decisions
Analyst / scribe	Crypto/rev/forensics, note discipline	Evidence template and screenshots	Floats to blocked lane, normalizes flags, builds timeline	Final PDF assembly and slide support

A five-person team is usually superior for finals because it reduces context switching and preserves documentation quality while two specialists keep attacking. The tradeoff is that it requires stronger leadership and more pre-practiced communication. ③⑩

Training roadmap, practice system, and prioritized resources

A realistic plan to win, not merely qualify, is a 24-week core program that you can compress to about 16 weeks for a four-month sprint or stretch to 48 weeks for a full-year program. The core idea is sequential: first build qualifier breadth, then specialize, then simulate the final. A 2025 UMS research project built specifically for GEMASTIK-style preparation found that a dedicated CTF plus Attack-Defense training system was functionally successful and scored well on usability testing, which is consistent with the practical need for structured internal practice rather than ad hoc study. ⁽⁴⁰⁾

Core 24-week roadmap

Weeks	Main goal	Weekly milestones	KPI by end of phase
1–4	Base environment	Linux CLI, Python scripting, Git, tmux, Docker basics, note-taking discipline	Stable local lab setup; 10 easy solves across categories; one reusable notes template
5–8	Web fundamentals	Burp workflow, auth/session labs, XSS, SQLi, SSRF, file upload, fuzzing	25–30 web labs solved; one script that automates a repeated web task
9–12	Crypto and reverse foundations	Modular arithmetic, RSA/AES basics, ECDSA nonce failures, Ghidra basics, assembly reading	20 crypto challenges and 8 reverse challenges solved; 3 clean write-ups
13–16	Pwn fundamentals	Stack overflows, ret2libc, PIE/NX/canary, format strings, basic heap and pwntools	15 pwn.college-style challenges solved; 5 exploit scripts from scratch
17–20	Forensics, stego, sniffing	PCAP triage, artifact extraction, metadata, browser traces, binwalk/zsteg, log reasoning	12 forensic/sniffing tasks solved in under 45 minutes each
21–24	Attack-Defense conversion	Weekly 4–6 hour scrim, patch-and-verify drill, exploit replay loops, timed write-up and presentation	Patch time below 20 minutes on familiar stacks; one complete mock final with PDF + presentation

How to compress to four months

If you only have four months, do not try to learn everything evenly. Compress the roadmap by merging weeks 1–4 into two heavy weeks, combining weeks 9–12 with weeks 17–20 into “support-category rotation,” and spending the final month almost entirely on web+pwn+defense+write-up execution. The likely winning specialization for a short cycle is one strong web player, one strong pwn/infra player, and one flexible analyst-scribe. ⁽⁴¹⁾

How to stretch to a full year

If you have 9–12 months, use the extra time for a proper pipeline: internal qualifiers, monthly scrimmages, alumni challenge review, and internal problem-setting. Teams that can set challenges usually become much better at identifying intended attack surfaces, patch boundaries, and presentation angles. That is one reason mature campuses keep reappearing in the podium table. ⁽⁴¹⁾

Prioritized resource ladder

The most useful study stack for GEMASTIK Cyber is surprisingly small if ordered correctly.

Priority	Resource	Use it for
Highest	, ,	Ground truth on format, timelines, topic

		families, rules, submission windows
Highest	''	Identify the real peer group and benchmark institutional competition
High	ctf-gemastik/penyisihan - https://github.com/ctf-gemastik/penyisihan and ctf-gemastik/penyisihan-2024 - https://github.com/ctf-gemastik/penyisihan-2024	Recover official/public qualifier source material where available
High	''''	Original problem-solving patterns and exploit/defense habits
High		Best single source for web repetition under realistic lab conditions
High		Best single structured path for pwn fundamentals and exploit iteration
High		Best structured track for crypto problem pattern recognition
Medium	CTFtime - https://ctftime.org	Competition calendar, archives, external benchmark events and write-ups

The reason these resources work together is alignment: official GEMASTIK sources define the contest you must win, while the lab platforms build the exact habits that public finalist write-ups repeatedly demonstrate.

⑫

Risks, common failure modes, and mitigation

The most common strategic failure is training for qualifiers only. Teams often get decent at Jeopardy-style solving, qualify, and then underperform badly in the final because they never practiced keeping a live service healthy while attacking others. The official rules make this format shift explicit every year, yet it remains the main competitive trap. The fix is simple: begin Attack-Defense scrimms at least a month before finals, even if your infra is crude. ⑬

The second failure mode is over-specialization without a scribe. A team with two killers and no disciplined writer often produces weak write-ups and presentations. GEMASTIK punishes that structure because the final includes same-day documentation and next-day presentation. The mitigation is to assign note ownership before the round starts and use a fixed evidence template. ⑭

The third failure mode is unsafe patching. In Attack-Defense, participants are tempted to “fix” a service by ripping out functionality, locking down too much, or introducing new crashes. That can kill uptime or break flag paths. The best mitigation is a three-step loop: patch minimally, run a health probe, and keep a rollback patch or container snapshot. ⑮

The fourth failure mode is manual exploitation. If a challenge can be solved once but not replayed automatically, it is much less valuable in a final. Public writing around GEMASTIK final challenges emphasizes exactly this point: the exploit should be scriptable fast enough to work inside a tick window or repeated scoring window. The mitigation is to require every finalist on offense to produce a replayable script, not just a one-off shell. ⑯

The fifth failure mode is rule violations under pressure. The official rules explicitly ban DoS, excessive damage, collaboration outside the team, and flag sharing. Teams that do not operationalize restraint can lose on procedure rather than technical quality. The mitigation is to appoint one captain responsible for escalation decisions and to log every risky action. ⑰

Open questions and limitations

Two parts of the public record remain incomplete. First, I did not find a fully official, publicly accessible, cross-year numeric dump of every 2023–2025 Cyber qualifier challenge and its exact point value, so the challenge-analysis section uses the strongest public metric available: official topic recurrence plus public archive/write-up confirmation. Second, I found strong official coverage for winners and finalist docs, but special-award detail such as 2025 Best Attacker was not as cleanly available from official pages in the sources I could verify, so I avoided presenting uncertain podium metadata as settled fact. ^[44]

The bottom-line conclusion is still robust: to win GEMASTIK Cyber, build a team that can qualify broadly, specialize decisively, automate relentlessly, defend conservatively, and explain clearly. In GEMASTIK, the champion is rarely just the smartest solver. The champion is the team with the best system. ^[45]

Sources

- [1] [8] [9] [10] [12] [13] [17] [19] [23] [24] [26] [43] [44] <https://pusatprestasinasional.kemendikdasmen.go.id/uploads/lampiran/Pedoman%20GEMASTIK%202023%20-%20revised%2020230509%20%281%29.pdf>
- <https://pusatprestasinasional.kemendikdasmen.go.id/uploads/lampiran/Pedoman%20GEMASTIK%202023%20-%20revised%2020230509%20%281%29.pdf>
- [2] [7] [14] [15] [41] [45] <https://gemastik.kemdiktisaintek.go.id/2023/peringkat/kompetisi>
- <https://gemastik.kemdiktisaintek.go.id/2023/peringkat/kompetisi>
- [3] [31] <https://nopedawn.github.io/posts/ctfs/2024/gemastik-17-xvii-qual-ctf-2024/>
- <https://nopedawn.github.io/posts/ctfs/2024/gemastik-17-xvii-qual-ctf-2024/>
- [4] [6] [11] [37] <https://gemastik.kemdiktisaintek.go.id/storage/unduh/buku-saku-gemastik-2024-562720.pdf>
- <https://gemastik.kemdiktisaintek.go.id/storage/unduh/buku-saku-gemastik-2024-562720.pdf>
- [5] [33] [40] <https://proceedings.ums.ac.id/korma/article/view/6724>
- <https://proceedings.ums.ac.id/korma/article/view/6724>
- [16] <https://gemastik.kemdiktisaintek.go.id/peringkat/kompetisi>
- <https://gemastik.kemdiktisaintek.go.id/peringkat/kompetisi>
- [18] [32] <https://riskyprsty.me/blog/Writeup-CTF-Gemastik-2024>
- <https://riskyprsty.me/blog/Writeup-CTF-Gemastik-2024>
- [20] <https://gemastik18.telkomuniversity.ac.id/dashboard/landing/peringkat-kompetisi>
- <https://gemastik18.telkomuniversity.ac.id/dashboard/landing/peringkat-kompetisi>
- [21] <https://cs.ui.ac.id/2025/11/03/fasilkom-ui-kembali-raih-juara-umum-gemastik-xviii-2025-dengan-sapu-bersih-9-penghargaan-bergengsi-4-emas-1-perak-1-perunggu-2-juara-harapan-1-best-paper-dan-1-best-defender/>
- <https://cs.ui.ac.id/2025/11/03/fasilkom-ui-kembali-raih-juara-umum-gemastik-xviii-2025-dengan-sapu-bersih-9-penghargaan-bergengsi-4-emas-1-perak-1-perunggu-2-juara-harapan-1-best-paper-dan-1-best-defender/>
- [22] <https://pusatprestasinasional.kemendikdasmen.go.id/uploads/lampiran/Pengumuman%20Finalis%20Gemastik%202023...pdf>
- <https://pusatprestasinasional.kemendikdasmen.go.id/uploads/lampiran/Pengumuman%20Finalis%20Gemastik%202023...pdf>
- [25] <https://patarisac.github.io/gemastik16-crypto-writeup/>
- <https://patarisac.github.io/gemastik16-crypto-writeup/>
- [27] [35] <https://zakigeyan.github.io/gemastik-2023/>
- <https://zakigeyan.github.io/gemastik-2023/>
- [28] [42] <https://portswigger.net/web-security/getting-started>
- <https://portswigger.net/web-security/getting-started>
- [29] <https://cryptohack.org/challenges/>
- <https://cryptohack.org/challenges/>
- [30] <https://pwn.college/intro-to-cybersecurity/binary-exploitation/>
- <https://pwn.college/intro-to-cybersecurity/binary-exploitation/>
- [34] [36] [38] [39] <https://static-gemastik18.telkomuniversity.ac.id/Panduan%20GEMASTIK%202025.pdf>
- <https://static-gemastik18.telkomuniversity.ac.id/Panduan%20GEMASTIK%202025.pdf>